Meet 0

October 19, 2026

Discrete

The answer key is below.

p	Answer	p	Answer
1	3	4	8
2	86	5	22
3	40	6	4435

1. Verity always tells the truth and Falsity always lies. A mystery number is given to both Verity and Falsity, and you are given that it is between 1 and 10, inclusive. Verity first says "The mystery number is a positive integer greater than 5"! Falsity then says "The mystery number is less than 8"! Given this, how many possible choices are there for the mystery number?

   Proposed by: Ahmed Ilyasov

   Answer: $\boxed{3}$

   Solution: Let the mystery number be x. We are given $1 \leq x \leq 10$. When Verity says the number is a positive integer greater than 5, it could be either 6, 7, 8, 9, or 10. Then, Falsity says $x < 8$, meaning $x \geq 8$. Thus, x could either be 8, 9, or 10, giving $\boxed{3}$ possibilities.

2. In AP Biology, the last ten test scores that Bob got were 100, 100, 100, 100, 100, 80, 70, 14, 14, and 2, in order. The mean of Bob's test scores is a, the median is b, and the mode is c. What is the mean of a, b, and c?

   Proposed by: Jeff Zhou

   Answer: $\boxed{86}$

   Solution: The mode is 100, as it appears 5 times. As there are ten total numbers, the median is the average of the middle two, which is $\frac{80+100}{2} = 90$. The mean is

   $$\frac{100+100+100+100+100+80+70+14+14+2}{10} = 68.$$

   Thus, the mean of a, b, and c is $\frac{68+90+100}{3} = \frac{258}{3} = \boxed{86}$.

3. The number 20 262 027 has exactly 4 positive integer factors. What are the last two digits of the sum of these factors?

   Proposed by: Ahmed Ilyasov

   Answer: $\boxed{40}$

   Solution: Note that $3 \mid 20262027$. Thus, $20262027 = 3 \cdot 6754009$. Now, suppose 6754009 is not a prime number. Consider categorizing all numbers with 4 factors. Either a number with 4 factors can be a perfect cube of a prime p, where the factors of that number are then 1, p, p^2, and p^3. Another possibility is that the number is a product of two prime numbers q and r, where the factors of that number are then 1, q, r, and qr. Since $6754009 \neq 3^2$, we must have that 6754009 is another prime.

   The sum of the four factors of 20262027 is then

   $$1 + 3 + 6754009 + 20262027 \equiv 1 + 3 + 9 + 27 = \boxed{40} \pmod{100}.$$

4. A pirate hides his 20 gold coins in four baskets, labelled 1, 2, 3, and 4. In baskets 1, 2, and 3 the amount of coins is a prime number. The amount of coins in basket 4 is more than the amounts of coins in baskets 1, 2, and 3 individually, but less than the combined amounts of coins in baskets 1, 2, and 3. If no two baskets have the same amount of coins, compute the amount of coins in basket 4.

   Proposed by: Ahmed Ilyasov

   Answer: $\boxed{8}$

Solution: Consider prime numbers under 20. They begin

$$2, 3, 5, 7, \ldots.$$

Now, we can test possibilities. If the amounts of coins in baskets 1, 2, and 3 were 2, 3, and 5 coins, then basket 4 would have 10 coins. However, note that 10 is not less than the combined amount of coins in baskets 1, 2, 3. We then try baskets with 2, 3, and 7 coins. Then, basket 4 would have 8 coins, which is greater than the amount of coins in every other individual basket and less than the combined total of the other three baskets. Thus, basket 4 has $\boxed{8}$ coins.

5. Kai and Darren each roll a fair 6-sided die, and the product of Kai and Darren's rolls is equal to D. Suppose there exists more than one possible value for the sum of Kai and Darren's rolls. Compute the sum of all possible values of D.

Note: A fair 6*-sided die is a die in which each of the* 6 *numbers* 1*,* 2*,* 3*,* 4*,* 5*, and* 6 *have an equal probability of being rolled.*

Proposed by: Ahmed Ilyasov

Answer: $\boxed{22}$

Solution: Note that if D is divisible by 5, there is only one possible value for the sum of Kai and Darren's rolls, as one of the rolls will equal 5 and the other will equal $\frac{D}{5}$.

Consider the case where D is divisible by 8. If one of the rolls is odd, then the other roll has to be divisible by 8, which is impossible. Thus, both rolls are even. If neither roll is equal to 4, then both of the rolls are either 2 or 6. However, since both 2 and 6 are divisible by 2 but not 4, any product of two of these integers is divisible by 4 but not 8. Thus, one of the rolls is equal to 4 and the other equal to $\frac{D}{4}$. This implies there is only 1 possible value for the sum of Darren and Kai's rolls.

Consider the case where D is divisible by 9. This means that both Darren and Kai either rolled a 3 or a 6; otherwise, if one of them rolled an integer not divisible by 3, the other would have to roll an integer divisible by 9, which is impossible. However, this implies that the product of Darren and Kai's rolls is equal to either 9, 18, or 36. There is only one possible value for the sum of Darren and Kai's rolls in each of these cases, so we disregard this case as well.

If D is a prime number or equal to 1, then obviously, there is only one possible value for the sum of Darren and Kai's rolls, as one of the rolls is equal to the prime itself and the other roll equal to 1.

Thus, D is a composite number not divisible by 5, 8, or 9. The possibilities are then $2 \cdot 2 = 4$, $2 \cdot 3 = 6$, and $3 \cdot 4 = 12$. Each of these possibilities work, as

$$2 \cdot 2 = 1 \cdot 4 = 4 \rightarrow 2 + 2 = 4,\ 1 + 4 = 5$$

$$2 \cdot 3 = 1 \cdot 6 = 6 \rightarrow 2 + 3 = 5,\ 1 + 6 = 7$$

$$3 \cdot 4 = 2 \cdot 6 = 12 \rightarrow 3 + 4 = 7,\ 2 + 6 = 8.$$

The sum of the possible values of D is then $4 + 6 + 12 = \boxed{22}$.

An alternative solution is to list the sample space of $6 \times 6 = 36$ dice products, which is feasible to do in contest and really does not take too long. Then, the desired D are the elements in the

sample space which appear more than twice, which are 4, 6, and 12, giving the answer of $\boxed{22}$.

	1	**2**	**3**	**4**	**5**	**6**
1	1	2	3	**4**	5	**6**
2	2	**4**	**6**	8	10	**12**
3	3	**6**	9	**12**	15	18
4	**4**	8	**12**	16	20	24
5	5	10	15	20	25	30
6	**6**	**12**	18	24	30	36

6. Minnetonka Math Team has 1 000 students, some of whom are friends with each other. Each student has at most 5 friends, and friendship is mutual.

   Each student starts with 1 dodecahedron. Every day, one student gives a dodecahedron to one of their friends. After some number of days, some student has all 1000 dodecahedrons. Compute the smallest possible number of days passed.

   Proposed by: Michael Luo

   Answer: $\boxed{4435}$

   Solution: Fix some student named Evan; we will transport all dodecahedrons to him. Let the *distance* between two students be the length of the shortest path of friendships between them. We want to minimize the sum of distances to Evan.

   First observe that at most one student has distance 0 to Evan; this is Evan himself. Now we claim the following.

   Claim. For all $n \geq 1$, there are at most $5 \cdot 4^{n-1}$ students who have distance n to Evan.

   Proof. We induct on n. The base case $n = 1$ holds since Evan has at most 5 friends. Now suppose the claim holds for some $n \geq 1$. Each student with distance $n+1$ from Evan is friends with some student with distance n from Evan. Meanwhile, each student with distance n from Evan is friends with at least 1 student with distance $n-1$ from Evan, therefore, each student with distance n from Evan is friends with at most 4 students with distance $n+1$ from Evan. Hence there are at most $4 \cdot 5 \cdot 4^{n-1} = 5 \cdot 4^n$ students with distance $n+1$ from Evan. □

   On the other hand, dropping the restriction that there are 1000 students, it is possible to have up to $5 \cdot 4^{n-1}$ students with distance n from Evan, by constructing a rooted tree with root Evan where Evan has five children and each non-Evan student has four children. Truncating the infinite tree, it is therefore optimal for there to be 1, 5, 20, 80, 320, and 574 students with distances 0, 1, 2, 3, 4, and 5 from Evan. Summing yields

$$1 \cdot 0 + 5 \cdot 1 + 20 \cdot 2 + 80 \cdot 3 + 320 \cdot 4 + 574 \cdot 5 = \boxed{4435}.$$